

ARVIN ARYANPOUR

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HIGHLIGHTS OF QUALIFICATIONS AND SKILLS

- Graduating software engineering student with hands-on experience building full-stack web applications, AI/CV systems, and microservices platforms; adept at delivering production-quality solutions in Python, FastAPI, React, and PostgreSQL.
- Strong proficiency in Python; strong experience in React and FastAPI; proficient in REST APIs, PostgreSQL, Git, and SQL; experienced with JavaScript, Java, C/C++, and C#; hands-on with PyTorch, YOLOv8, scikit-learn, and Unity ML-Agents for AI and ML development.
- Built end-to-end AI inference pipelines, full-stack analytics platforms, and scalable microservices — with demonstrated results: 15% API performance gain, 65%–80% test coverage improvement, and 200ms load time reduction in production.
- Experienced in Agile/Scrum workflows, code reviews, automated testing, and iterative software delivery; strong understanding of software engineering best practices and clean, maintainable code.
- Proven communicator and collaborative team leader, developed through 10+ years competing and coaching in competitive soccer and 5+ years in volleyball; brings the same adaptability, accountability, and team-first mindset to engineering teams and cross-functional collaboration.
- Fast, self-directed learner who consistently picks up new technologies and applies them to real problems; independently integrated PyTorch, YOLOv8, and Unity ML-Agents into personal and academic projects without formal coursework.

EXPERIENCE

Web Developer

Dec 2024 – Present

Pro Insulation Contracting

Toronto, ON

- Built lead-generation web applications in FastAPI with REST APIs, improving conversion rates and user engagement.
- Optimized performance and SEO, reducing page load times and increasing inbound traffic.
- Integrated analytics pipelines to track user behaviour and marketing-performance metrics, enabling data-driven product and campaign decisions.
- Collaborated directly with stakeholders to translate business requirements into scalable full-stack solutions, ensuring on-time delivery of production-ready features.

Software Engineer Intern

Apr 2023 – Oct 2023

VerticalScope

Toronto, ON

- Developed 5+ reusable React UI components deployed across 1,200+ high-traffic community websites, improving frontend consistency and enabling faster platform-wide feature rollout.
- Achieved a 200ms reduction in homepage load times, boosting user retention and ad impression metrics across the platform.
- Optimized backend API response times by 15% through strategic database indexing and query refactoring, reducing server-side latency for all active users.
- Increased codebase test coverage from 65% to 80% by authoring unit and integration tests, preventing critical regressions across the production environment.

Footwear Sales Associate

Sep 2024 – Present

Sporting Life

Toronto, ON

- Consistently earned the #1 sales ranking in my department by developing targeted product recommendations based on individual customer needs and purchasing patterns, converting consultative interactions into high-value purchases.
- Exceeded Google review targets by up to 500% through consistently exceptional customer service and follow-through.
- Adapted sales strategies in a high-volume retail environment based on direct customer feedback, consistently translating technical product knowledge into confident purchase decisions and exceeding weekly KPIs.

PROJECTS

Real-Time Volleyball Analytics Platform

- Built a full-stack web application to analyze match footage and track player and ball movement in real time.
- Engineered a YOLOv8 + ByteTrack computer vision pipeline achieving 87.9% detection precision for multi-object tracking.
- Developed a FastAPI + React interface for live inference, frame-level analytics, and interactive match review.

Smart Parking Microservices System

- Designed a backend platform managing real-time space allocation and dynamic pricing for smart parking lots.
- Engineered a microservices architecture with authenticated REST APIs across independent services and PostgreSQL persistence.
- Defined clean inter-service contracts to support horizontal scalability across the platform.

Reinforcement Learning Agent – Capstone (Unity ML-Agents)

- Built a 3D video game where the player's AI companion NPC uses reinforcement learning to navigate and support the player in real time.
- Designed and trained a PPO agent with custom reward shaping and observation-space design.
- Instrumented the full training loop to validate convergence and tune agent behaviour.

EDUCATION

B.Eng., Computer Software Engineering (Honours)

Toronto Metropolitan University

2021 – 2026

Toronto, ON

EXTRACURRICULARS

MUES Engineering Council – Event Coordinator

Planned and coordinated faculty-wide engineering events as part of the Metropolitan University Engineering Society.

TMU Formula Racing Team – Software Team

Contributed to embedded software and data systems on TMU's student-built formula race car across two competitive seasons.

Competitive Athletics – Soccer (10+ years) and Volleyball (5+ years)

Competed and coached in soccer and volleyball at a competitive level, developing leadership, communication, and resilience across high-pressure team environments.